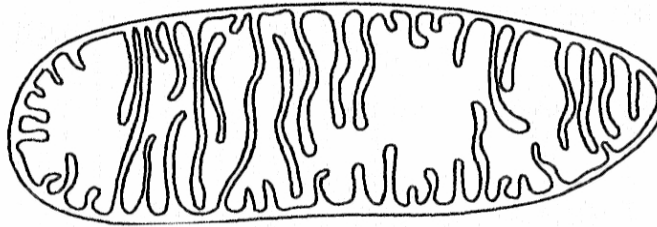


Answer **ALL** questions

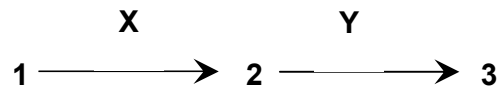
- 1 When a glycoprotein is being synthesized for secretion from a cell, which route is it most likely to take?
- A golgi apparatus → rough endoplasmic reticulum → smooth endoplasmic reticulum
 - B rough endoplasmic reticulum → golgi apparatus → smooth endoplasmic reticulum
 - C smooth endoplasmic reticulum → rough endoplasmic reticulum → golgi apparatus
 - D rough endoplasmic reticulum → smooth endoplasmic reticulum → golgi apparatus
- 2 The diagram shows an organelle found in a plant cell.



Which molecules are found in this organelle?

- A cellulose and protein
 - B phospholipid and protein
 - C triglyceride and glucose
 - D starch and glucose
- 3 Which one of the following statements is **INCORRECT**?
- A There are 3 polypeptide chains in the collagen helix, and only one in the α -helix.
 - B Proline is a major component of collagen but not of the α -helix.
 - C Every third residue in the α -helix must be glycine, but there is no sequence regularity in collagen.
 - D Some hydrogen bonding exists between different chains in collagen, but hydrogen bonding occurs within the same chain in the α -helix.

- 4 In the diagram below, the numbers 1 – 3 refer to different amino acids and the letters X and Y refer to different enzymes.



The artificial introduction into the cell of an excess of amino acid 3 reduces the rate of reaction catalysed by enzyme Y. What is the cause of this?

- A Competitive inhibition
 - B End-product inhibition
 - C Positive feedback
 - D Enzyme denaturation
- 5 Succinic dehydrogenase is the enzyme that catalyses the oxidation of succinic acid during cellular respiration. If malonic acid is added to the system, the rate of reaction is reduced.
- An increase in substrate concentration will, however, increase the rate of reaction again. From the above information, what can you conclude about malonic acid?
- A It binds to the allosteric site of the enzyme.
 - B It increases the pH of the system.
 - C It has a similar molecular configuration to that of succinic acid.
 - D It forms a permanent attachment to the active site of the enzyme.
- 6 At which stage of meiosis does each of these events occur?

	Crossing over	Independent assortment of the bivalents	Separation of sister chromatids
A	Metaphase I	Prophase I	Anaphase I
B	Metaphase I	Prophase I	Anaphase II
C	Prophase I	Metaphase I	Anaphase I
D	Prophase I	Metaphase I	Anaphase II

- 7 The figure below shows a stage in the division of a cell from an anther of a flower.



Identify the type and stage of nuclear division shown, as well as the correct diploid number of the plant.

	Type of nuclear division	Stage of nuclear division	Diploid number
A	Mitosis	Anaphase	14
B	Meiosis	Anaphase I	14
C	Meiosis	Anaphase II	28
D	Meiosis	Anaphase II	14

- 8 Which of the following enzymes or proteins is **NOT** used in the process of DNA replication?
- A** Primase
 - B** DNA polymerase
 - C** Translocase
 - D** Single-strand binding protein
- 9 How many different polypeptides, each consisting of 3 amino acids, can be made if the number of different amino acids available is 10?
- A** 30
 - B** 1000
 - C** 3000
 - D** 59049

10 The following events occur in the extension of telomere:

1. Further extension of 3' end of telomere
2. Translocation of telomerase to end of telomere
3. Reverse transcription to add to telomere sequence
4. Complementary base-pairing of RNA template with telomere overhang
5. Synthesis of complementary C-rich strand to form double stranded DNA

In which order do these events take place?

- A** 3 2 4 5 1
B 3 4 2 1 5
C 4 2 3 5 1
D 4 3 2 1 5

11 Lactose is a disaccharide that can be used by *E. coli* as source of energy instead of glucose. Genes involved in lactose metabolism are called *lacZ*, *lacY* and *lacA*. In wild type bacteria, *lacZ*, *lacY* and *lacA* expression rises dramatically when the lactose concentration rises and glucose concentration falls.

A mutant strain of *E. coli*, abnormal in its use of lactose was identified. Tables **A** and **B** show the expression of each protein by *E. coli* when grown in the absence and presence of lactose. Assume lactose was the only carbon source.

(✓ = protein is expressed; X = protein is not expressed)

Table A

	Without lactose		
<i>E. coli</i> strain	<i>lac Z</i>	<i>lac Y</i>	<i>lac A</i>
Wild type	X	X	X
Mutant strain	X	X	X

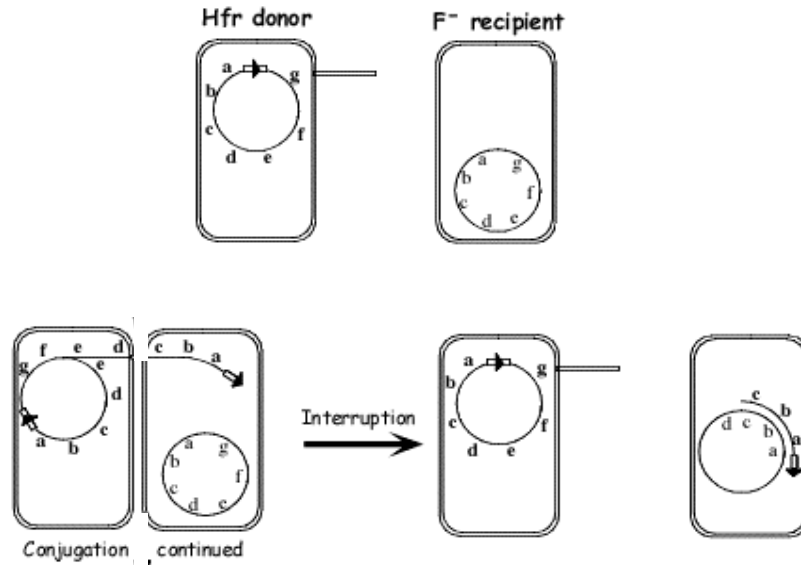
Table B

	With lactose		
<i>E. coli</i> strain	<i>lac Z</i>	<i>lac Y</i>	<i>lac A</i>
Wild type	✓	✓	✓
Mutant strain	✓	X	X

Which of the following best accounts for the observed data?

- A** Deletion of *lacY* coding sequence
B Presence of premature stop codon in *lacZ* gene
C Deletion of the entire lactose operon
D Deletion of *lacZ* coding sequence

12 The diagram shows the conjugation process of an Hfr cell with an F⁻ cell.



	Genes replaced	Process of replacing genes	Recipient cell after conjugation
A	a, b, c	chiasmata	Hfr
B	a, b, c	crossing over	F ⁻
C	d, e, f	chiasmata	Hfr
D	d, e, f	crossing over	F ⁺

13 Generalized transduction is distinguishable from specialized transduction by the fact that

- A** generalized transduction may be used to move any gene, whereas specialized transduction moves only certain genes.
- B** selective medium is required for generalised transduction, whereas selective medium is not required for specialized transduction.
- C** donor DNA must be purified from the donor for generalised transduction, whereas specialized transduction involves movement of DNA by phages.
- D** generalized transduction is possible in generally all organisms, whereas specialized transduction is possible only in special groups of organisms.

- 14 Viruses are classified according to several characteristics, which includes all **EXCEPT**
- A DNA or RNA as the genetic material.
 - B single or double-stranded nucleic acids.
 - C circular or linear nucleic acid.
 - D presence or absence of metabolic machinery.
- 15 An mRNA is 336 nucleotides long, including the initiator and termination codons.
- The number of amino acids in the protein translated from this mRNA is
- A 110
 - B 111
 - C 112
 - D 330
- 16 Which of the following statements is **INCORRECT**?
- A Initiation of transcription requires binding of RNA polymerase to a promoter.
 - B All primary transcripts (RNA) are processed before being used for translation into protein.
 - C Double-stranded DNA, not single-stranded DNA, is used for transcription by RNA polymerase.
 - D Uracil, not thymine, is incorporated into RNA during transcription.
- 17 Which of the following is a method of post-translational control in eukaryotic cells?
- A Selectively degrading the proteins when they are transported to targeted destinations.
 - B Preventing the attachment of ribosomes.
 - C Inactivates essential translational factors.
 - D Selectively degrading the mRNA transcripts.

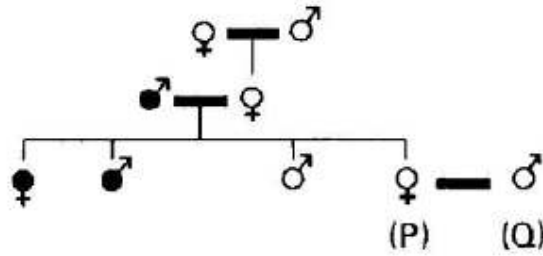
- 18** Which of the following shows the possible effects of a single nucleotide substitution in each of the following locations in a gene on the production of the protein product of that gene?

	Promoter	Terminator	Start Codon	Stop Codon	Middle of an intron
A	No protein product is formed	Protein product is shorter than normal	Protein product is longer than normal	Protein product is normal	Too much protein product is produced
B	Too much protein product is produced	Protein product is normal	No protein product is formed	Protein product is longer than normal	Protein product is normal
C	Protein product is normal	Protein product is longer than normal	Protein product is shorter than normal	Too much protein product is produced	Protein product is longer than normal
D	Protein product is longer than normal	Too much protein product is produced	Protein product is normal	Protein product is shorter than normal	No protein product is formed

- 19** Which statement is **NOT** true about proto-oncogenes?

- A** Proto-oncogenes encode proteins which promote the cell cycle.
- B** Proto-oncogenes and tumor suppressor genes work the same way.
- C** When the expression of proto-oncogenes becomes aberrant, they can become oncogenes.
- D** At least one copy of a proto-oncogene needs to be mutated to lead to a malignancy.

- 20 In a species where the female is homogametic, a sex-linked allelic pair of genes controls pigmentation. The following results were obtained during the course of a breeding experiment.



  = Individuals with normal pigmentation
  = Individuals with abnormal pigmentation
 = A 'mating line'

Which one of the following ratios of offspring will be produced when **P** and **Q** are bred together?

- A 2 females (both normal): 2 males (both abnormal)
 B 2 females (both carriers): 2 males (both normal)
 C 2 females (1 carrier, 1 abnormal): 2 males (1 normal, 1 abnormal)
 D 2 females (1 carrier, 1 normal): 2 males (1 normal, 1 abnormal)
- 21 Skin colour in onions is controlled by two pairs of alleles, **Ss** and **Rr**, which segregates independently. The allele **S** is dominant and must be present to allow development of pigment in the skin. In its absence, the onion skin is white. Allele **R** is dominant and gives a red colour; the recessive **r** gives a yellow colour.

What is the ratio of phenotypes in the offspring of a cross between plants of genotypes **SsRR** and **ssrr**?

- A All red
 B 1 red : 1 white
 C 3 red : 1 white
 D 1 white : 2 red : 1 yellow

- 22** A strain of toad has only one nucleolus in the nucleus of each cell instead of the usual two. When toads with one nucleolus per cell are mated, approximately a quarter of the offspring have two nucleoli per nucleus, half have one nucleolus per nucleus and a quarter have no nucleoli.

What is the most likely explanation of these results?

- A** The possession of one nucleolus is due to autosomal linkage.
- B** The possession of one nucleolus is due to the heterozygous condition.
- C** The allele for the presence of two nucleoli is recessive.
- D** The allele for the presence of two nucleoli is dominant.

- 23** In mice, two genes code for hair colour and length. These genes assort independently. The alleles are represented below:

Gene for hair colour: **B** black
 b brown

Gene for hair length: **L** short
 l long

A mouse with short, black hair was crossed with a mouse with long, brown hair. Half of the resulting offspring had short hair and half of the offspring had black hair. The genotype of the parent mouse with short, black hair must have been

- A** LLBB.
 - B** LLBb.
 - C** LIBB.
 - D** LIBb.
- 24** A zygote possesses a $(2n - 1)$ chromosome number. This means that one of the gametes involved in its formation was a product of
- A** translocation during meiosis.
 - B** crossing over during meiosis.
 - C** non-disjunction during meiosis.
 - D** deletion of a gene during meiosis.

- 25** Dominique has never had much energy. He goes to a doctor for help and is sent to the hospital for some tests. There they discover his mitochondria can use only fatty acids and amino acids for respiration, and his cells produce more lactate than normal.

Which of the following is the best explanation for his condition?

- A** His mitochondria lack the transport protein that moves pyruvate across the inner mitochondrial membrane.
 - B** His cells cannot move NADH from glycolysis into the mitochondria.
 - C** His cells contain an inhibitor that prevents oxygen use in his mitochondria.
 - D** His cells have a defective electron transport chain, so glucose is reduced to lactate instead of to acetyl CoA.
- 26** Which statement on chemiosmotic mechanism of the coupling of substrate oxidation to the phosphorylation of ADP is **INCORRECT**?
- A** The main protein complexes of the electron transport chain in mitochondria pump protons out into the intermembranal space.
 - B** A central role is played by a proton gradient that exists across the inner mitochondrial membrane.
 - C** The pH inside the mitochondrial matrix is less than that in the cell cytoplasm.
 - D** The electrochemical 'force' which drives the synthesis of ATP is called the proton-motive force.
- 27** The weedkiller DCMU blocks the flow of electrons chains in photophosphorylation. Why does this kill the plant?
- A** ATP and reduced NADP are not produced.
 - B** ATP and reduced NAD are not produced.
 - C** Photoactivation of the chlorophyll cannot occur.
 - D** Photolysis of water does not occur.

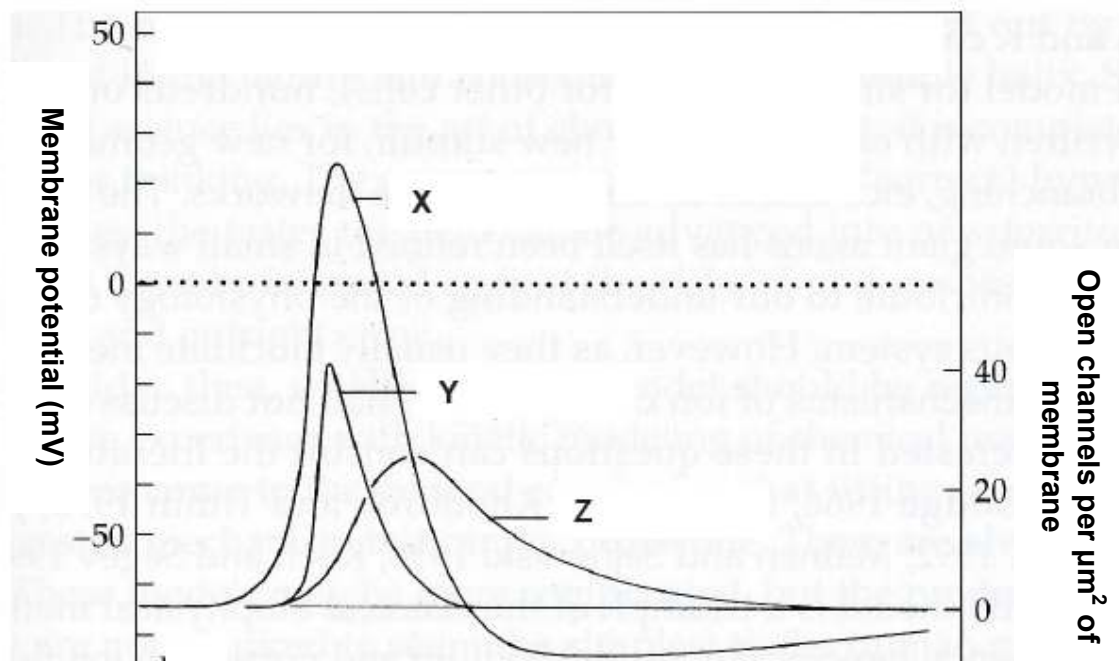
28 Curare causes paralysis of striated muscle. When curare is applied to nerve muscle preparation:

- (i) the muscle does not contract when the nerve is stimulated, but
- (ii) if a stimulus is applied directly to the muscle, contraction occurs.

When curare is applied only to the nerve, the nerve is stimulated. Which of the following correctly explains the effect of curare?

- A** It inhibits the development of an action potential in the nerve fibre.
- B** It stimulates the formation of acetylcholine at the nerve cell end-plate.
- C** It prevents the depolarization of the post-synaptic membrane.
- D** It affects the formation of the enzyme cholinesterase.

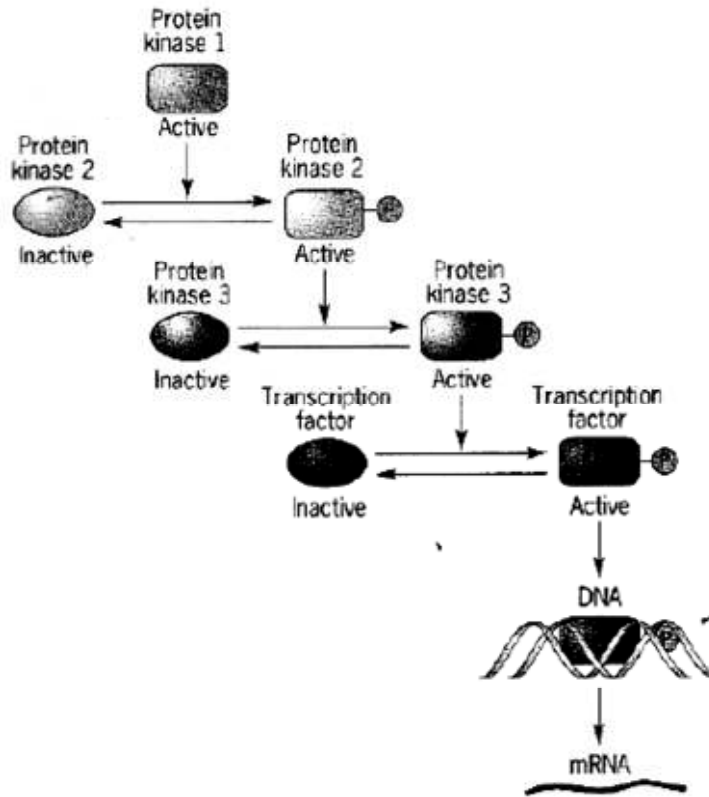
29 What do **X**, **Y** and **Z** in the above figure of an action potential represent?



	X	Y	Z
A	Change in membrane potential	Na ⁺ permeability	Resting membrane potential
B	Na ⁺ permeability	Change in membrane potential	Ca ²⁺ permeability
C	Change in membrane potential	Na ⁺ permeability	K ⁺ permeability
D	Resting membrane potential	Ca ²⁺ permeability	K ⁺ permeability

- 30** The figure below shows part of a signalling pathway in a cell. The pathway leads to the activation of a gene that codes for a molecular inhibitor of a cyclin-dependent kinase responsible for moving a cell into the S phase of the cell cycle.

How would a mutation that results in a non-functional Protein kinase 3 affect the cell's rate of division?



- A** Cell division would proceed at a rapid rate.
- B** Cell division would proceed at the normal rate.
- C** Cell division would proceed, but at a decreased rate.
- D** Cell division would be completely inhibited.
- 31** Which of the following describes correctly the action of G-protein?
- A** Intracellular signal molecule activates G-protein resulting in a change in surface configuration.
- B** G-protein acts as GTPase enzyme and hydrolyses GDP to GTP.
- C** G-protein linked receptor is inactivated when the activated G-protein binds to a membrane protein.
- D** G-protein mechanism results in multiple cellular responses.

32 Which sequence of events correctly describes evolution?

- I Differential reproduction of the spiders occurs.
- II A new selection pressure occurs.
- III Allele frequencies within the spider population change.
- IV Poorly adapted spiders have decreased survivorship.

A II, IV, I, III

B II, IV, III, I

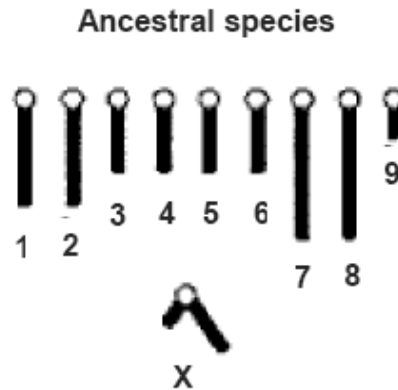
C IV, I, III, II

D IV, III, I, II

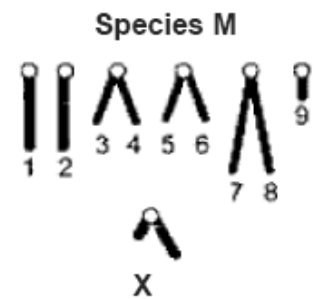
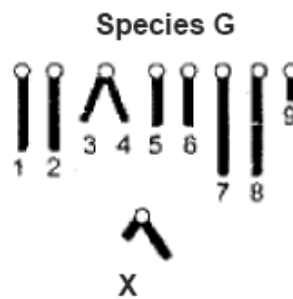
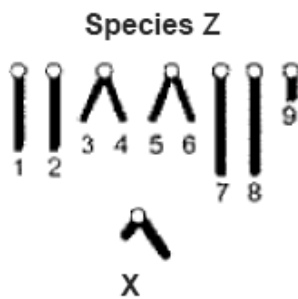
33 Modifications in the organisation of the basic pentadactyl limb (having five limbs) structure found in invertebrates provides good evidence for the principle of

- A** adaptive radiation.
- B** genetic drift.
- C** convergent evolution.
- D** inheritance of acquired characters.

- 34 In some Australian insects, new species have arisen through changes that occurred to chromosomes in an ancestral species. Such changes may involve the joining together of chromosomes, the loss of whole or parts of chromosomes, and rearrangement of the genetic material within chromosomes. One ancestral species has the following haploid set of chromosomes.



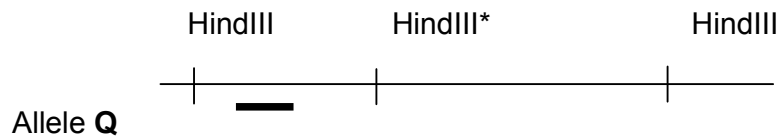
As the changes in chromosomes accumulate, a number of different species can result from a single ancestral species. Three species that have evolved from the ancestral species shown above have the haploid sets of chromosomes shown below.



The most likely order of evolution of these species is

- A ancestral species, species Z, species G, species M.
 - B ancestral species, species G, species M, species Z.
 - C ancestral species, species M, species G, species Z.
 - D ancestral species, species G, species Z, species M.
- 35 Which of the following statements could **NOT** be used to describe a species?
- A A group of organisms showing distinctly similar autosomes.
 - B A group of organisms showing analogous body structures.
 - C A group of organisms capable of mating to produce viable offspring.
 - D A group of organisms sharing the same ecological niche.

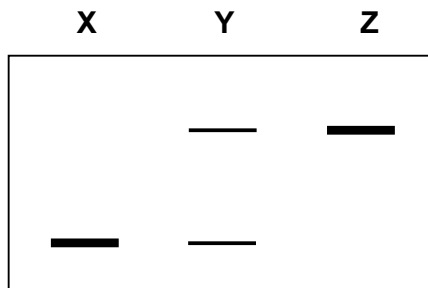
- 36 Which of the following correctly explains why it is more advantageous to use yeast cells rather than bacterial cells for expression of eukaryotic DNA?
- A Yeast cells can be cultured more readily than bacterial cells.
 - B Yeast cells can excise introns from the RNA transcript, but bacterial cells cannot.
 - C Yeast cells can reproduce more rapidly than bacterial cells.
 - D Yeast cells can produce proteins at a faster rate than bacterial cells.
- 37 The region of the genome containing the RFLP used in this analysis is shown below.



HindIII indicates the restriction sites for this enzyme and * indicates the polymorphic site which is missing in the recessive allele **q**.

The black bar indicates the position of the probe used to detect the RFLP.

DNA fragments from three different individuals, **X**, **Y** and **Z**, were subjected to restriction digestion by HindIII and separate by gel electrophoresis. The following results were obtained.



Deduce the genotype of individual **Z**.

- A QQ
- B Qq
- C qq
- D X^QY

- 38 Which of the following statements is **FALSE** about hematopoietic stem cells?
- A Hematopoietic stem cells can differentiate into all specialised cells.
 - B Hematopoietic stem cells can differentiate into all blood cell types.
 - C Hematopoietic stem cells can be found in the bone marrow.
 - D Hematopoietic stem cells are able to reproduce continually.
- 39 Which of the following disorders would gene therapy be **LEAST** effective?
- A Cystic fibrosis
 - B Huntington's disease
 - C Sickle cell anaemia
 - D Type II diabetes
- 40 Plants are more readily manipulated by genetic engineering than are animal cells because
- A plant genes do not contain introns.
 - B more vectors are available for transferring recombinant DNA into plant cells.
 - C a somatic plant cell can often give rise to a complete plant.
 - D plant cells have larger nuclei.

End of H2 Paper 1